Case 1: Sales Performance Analysis

Company Background

TechSupply Co. is a regional distributor that sells electronics, office supplies, and office furniture to both retail and business customers. Management wants to better understand sales trends, customer purchasing behavior, and regional performance to improve future sales strategies.

You have been hired as a junior data analyst to explore the sales dataset and provide insights. Your task is to use Excel filtering and sorting tools to answer business questions and provide recommendations.

Dataset: use the *techsupply_dataset.xlsx*

Dataset Features

The dataset simulates a **technology supply company sales system** and includes:

Column	Description
Order_ID	Unique order number
Date	Transaction date
Region	East, West, North, South
Sales_Rep	Assigned sales representative
Product_Category	Electronics, Furniture, Office Supplies
Product	Individual product sold
Units_Sold	Quantity sold
Unit_Price	Price per unit
Total_Sales	Total revenue for the order
Customer_Type	Business or Retail
Payment_Method	Credit Card, Debit, Cash, Wire

Complete the following 5 tasks.

REGIONAL PERFORMANCE ANALYSIS

Task 1: Use filters to explore sales by region.

1. Filter by Region = East

2. Sort Total_Sales from Largest to Smallest

Questions:

1. What is the largest sale in the East region?
   -Laptop
2. Which product category appears most frequently?
   -Office supply
3. Which sales representative appears most often?

   -Brown

CUSTOMER BEHAVIOR ANALYSIS

Task 2: Filter by

1. Customer_Type = Business

Questions

1. Which product category do business customers purchase most often?
   -Mode for business customer purchase is Office supply

2. What is the largest business transaction?

-Order 2290, 2026-03-20, South, Singh, Electronics, Laptop 76 unit, Unite price 837.45, Total sales of 63,646.20 paid by cash

3. Which sales rep handles the most business orders?

-Mode for sales rep that is filtered by business orders is Brown

PRODUCT PERFORMANCE

Task 3: Use filters to analyze **Product_Category**.

Analyze each category separately:

- Electronics
- Furniture
- Office Supplies

Questions

1. Which category appears to generate the highest total sales?

-Furniture making 1,290,860.62 of total sales

2. Which specific product appears most frequently?
   -Filling cabinet appeared the most frequently
3. Which category tends to have the largest individual transactions?

   -All three categories have the same amount of individual transactions which is 1.

PAYMENT PATTERN ANALYSIS

Task 4: Filter by Payment_Method

Analyze:

- Credit Card
- Debit
- Cash
- Wire

Questions

1. Which payment method is used most frequently?
   -Credit cart appeared the most payment method
2. Do business customers prefer a specific payment type?
   -Debit card
3. Which payment type is associated with larger purchases?

   -Cash

TOP SALES REPRESENTATIVES

Task 5: Sort by:

1. Sales_Rep
2. Total_Sales(Largset > Smallest)

Questions

1. Which sales rep appears to handle high-value transactions?
   -Singh handled the most transactions that totals to 20,000 and up for total sales.
2. Which rep appears most frequently in the dataset?
   -Brown appears most frequently in the dataset
3. Which region do they appear to sell in most often?

   -Brown appear to sell the most in West while Singh appear to sell the most in South

ANALYST RECOMMENDATION

Imagine you are presenting your findings to management.

Answer the following:

1. Which region appears to be the strongest market?
   East appeared the most in the sets which tells us that they provide more sales than the other regions
2. Which product category should the company promote more?
   Companies should promote headphones more because they are the product that has the least sales.
3. Are business customers or retail customers more valuable?
   Retail customers have the highest total sales compared to business customers therefore, Retail customers provide more value to the company
4. What two recommendations would you give the sales manager?

   -Brown and Singh appear the most across the dataset, it would be helpful if a company were to create a mentorship program where these 2 leading the program to help other sales representatives with selling.

   -Focusing more on promoting products that are the lowest in sales like Headphones.

Bonus

Build

1. Bar Chart – Sales by Product Category
2. Pie Chart – Customer Type Distribution
3. Column Chart – Sales by Region

What story does the data tell?